Week 04 Artificial Intelligence

Documentation

Step 1: Dataset Choice

I selected a dataset on youth wellbeing and mental health in Aotearoa New Zealand from Stats NZ, concentrating on metrics like stress levels, life satisfaction, and screen time patterns. I chose this dataset since it directly relates to the emotions I've documented when using my phone. It made it possible for me to contrast my own experience with more general patterns among New Zealand's youth.

Step 2: Understanding the Data

I investigated the dataset's structure after uploading the CSV into an AI program. The information contained columns like: Age range, Gender, stress levels reported, hours spent on screens, scores for wellbeing

The information revealed trends in young people's stress experiences as well as potential relationships between lifestyle factors like screen usage and wellbeing. I discovered from my conversation with the AI that the information is made up of aggregated survey responses, which means it reflects broad patterns rather than unique experiences.

This data points to a number of potential narratives. It can demonstrate, for instance, how more screen time may be linked to increased stress or worse wellness. It also calls into question the potential effects of habits, such as constantly checking phones, on emotional states. But there are restrictions as well. The dataset does not capture context, such as why people feel stressed or what they are doing on their devices. Depending on who answered the poll and how the questions were understood, bias might potentially exist.

Step 3: Multiple Representations

At first, the AI produced a simple bar graph that displayed stress levels by age group. This was fairly general, but it was straightforward. It didn't really relate to my topic of emotions and made use of default colors.

I then instructed the AI to use emotion-based images, akin to my previous doodles, to remodel the data. Screen time was mapped to size, and stress levels were indicated by color intensity (e.g., red for severe stress). This clearly connected the data to my own recorded emotions, such as fatigue and tension, making it feel more relatable and real.

Lastly, I came up with an idea for an interactive p5.js visualization in which:

Every individual or group turns into a moving circle. Stress increases movement speed and chaos. Larger size = more screen time

Instead of using static statistics, this version illustrates how stress may feel overpowering and dynamic.

Step 4: Critical Evaluation

At first, the AI produced expected results, such as bar charts and straightforward layouts. It prioritized clarity over emotional effect and assumed an impartial, academic audience. To make it more expressive and related to my idea, I had to direct it.

The AI's fundamental premise was that data ought to be delivered impartially and without bias. My experiment, however, contradicts this by demonstrating how emotional data may be presented in more intimate and captivating ways.

The interactive version, which captures both the data and the sentiment behind the data, was the most fascinating portrayal. Instead of only reading patterns, it enables viewers to experience them.

I would spend less time on conventional chart styles and more time exploring original visual metaphors if I were designing this without AI. AI was helpful for producing ideas rapidly, but it needed clear guidance to go beyond general outcomes.

Reflection

I selected a Stats NZ dataset on stress and youth wellness in Aotearoa New Zealand for this project. I chose this dataset since it directly relates to my previous experiment in which I monitored my own feelings when I picked up my phone. I aimed to investigate the connection between individual experience and more general social trends by merging my personal data with national-level data.

I was able to rapidly comprehend the dataset's structure with the use of AI techniques. I was able to inquire about the meaning of each column, the number of entries, and potential patterns by uploading the CSV. This makes it simpler to pinpoint important factors like screen time and stress levels. Nevertheless, the AI frequently defaulted to generic interpretations, such as proposing straightforward linkages without considering the context, and tended to concentrate on superficial descriptions. It did not critically examine how the data might be biased, who gathered the data, or whose viewpoints were included or excluded.

I had to make thoughtful design choices while leading the AI in order to go beyond typical results. At first, the AI generated simple bar charts with muted colors that had nothing to do with my idea. I pushed it in the direction of more expressive representations, including converting emotions into animated forms and mapping stress to color intensity. Through this process, I discovered that AI frequently leans toward convention and clarity and needs strong supervision to generate more creative or meaningful solutions.

Creating multiple representations of the same data showed how much interpretation affects understanding. A bar chart emphasises comparison and clarity, while my emotion-based and interactive p5.js visualisations emphasise feeling, repetition, and experience. The interactive version, in particular, allows viewers to sense patterns over time, such as the repetition of "tired" in my own data, which connects to broader stress trends. This demonstrates that data is not neutral—it can be framed in ways that highlight different narratives.

Reading:

I was prompted to consider power in data more critically by the concepts presented in Data Feminism by Lauren F. Klein and Catherine D'Ignazio. Their claim that systems of inequality shape data caused me to wonder which experiences are included in the dataset and which are not. Although my own data offers a unique viewpoint, it is still constrained and subjective. This emphasizes how important it is to take into account different points of view instead than interpreting data as objective truth.

In a similar vein, Kirikowhai Mikaere's explanation of Māori data sovereignty changed my perspective on data as a resource. She emphasizes that data is a strategic asset that Māori should control for Māori development, not just information. This led me to wonder who owned the dataset I utilized and how it was being used. It also highlighted the necessity for culturally grounded approaches to data, as national databases might not accurately represent Māori viewpoints or interests.

In general, using AI as a design tool proved both beneficial and restrictive. It accelerated research and idea generation, but in order to prevent generic results, it required guidance and critical thought. If I had more time, I would explore more culturally sensitive approaches to depict wellbeing and add real-time data to my project. I've learned from this endeavor that data is about people, power, and perspective in addition to numbers.

Difficulties:

Working with new tools and software, particularly coding environments like Visual Studio Code, was one of the biggest challenges I encountered during this project. I had little prior coding experience, so it was difficult for me to comprehend how everything works together, including properly configuring files, executing code, and troubleshooting difficulties. It was annoying and time-consuming since even minor errors, like missing brackets or improper syntax, may prevent the application from functioning at all.

I was still learning the fundamentals, so I spent a lot of time trying to understand how the code operated instead of coming up with more complex or original ideas. Because I had to keep my ideas quite minimal to guarantee they worked properly, this restricted the range of what I could create. Even when I had a clear notion in mind, there were moments when it felt challenging to put my thoughts into code.

I had to continually alternate between learning the tool and finishing the task when using new applications. Compared to using tools I am already familiar with, this slowed down my workflow and reduced process efficiency. Nevertheless, in spite of these difficulties, the experience helped me develop fundamental coding and problem-solving skills, and I gradually gained confidence.

Ultimately, even though my inexperience limited the project's technical potential, it also forced me to pick up new abilities and adapt, which is crucial for the design process.